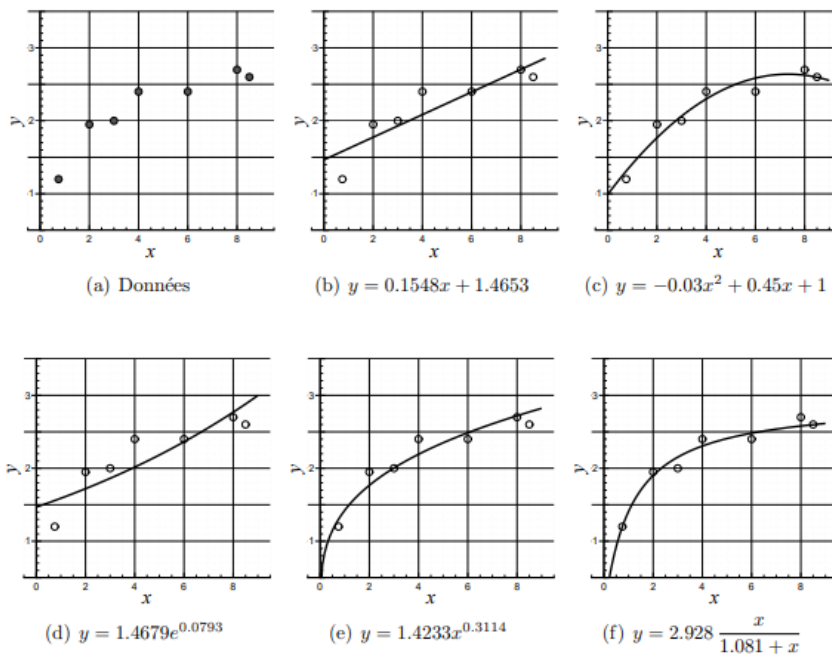


- Approximation in the sense of Least Squares -

Exercise 1: Discrete Least squares approximation

We have the experimental data in the table below. If the least squares method is used to approximate these data by (b) a straight line, (c) a parabola, (d) an exponential model $y = \alpha e^{\beta x}$, (e) a power law $y = \alpha x^{\beta}$, or (f) a saturated growth rate model $y = \frac{\alpha x}{\beta + x}$, the curves shown in the figure below are obtained.

x	0.75	2	3	4	6	8	8.5
y	1.2	1.95	2	2.4	2.4	2.7	2.6



- (1) In your opinion, which of the five approximations obtained is the best?
- (2) How can one determine with certainty the best approximation?
- (3) Determine the best linear approximation of the given data using least square approximation.
- (4) Provide the best quadratic approximation of the given data using least square approximation.

Exercise 2: Linear Least Squares approximation: Problem 1

The average number g of goals per game scored by a hockey player seems to be related linearly to two factors: the number x_1 of years of experience and the number x_2 of goals in the preceding 10 games. The data on the following table were collected on four players. Find the linear function $g = a_0 + a_1x_1 + a_2x_2$ that best fits these data

x_1	5	3	1	2
x_2	3	4	5	1
g	0.8	0.8	0.6	0.4

Exercise 3: Linear Least Squares approximation: Sales forecast

Mionneur is a company that manufactures and sells spare parts for trucks, catering to road professionals. The company's manager wants to estimate the sales revenue for the year $N + 1$. He provides the sales revenue data for the past seven years:

Year	Sales Revenue (in units)
N	4100
$N - 1$	3850
$N - 2$	3800
$N - 3$	3450
$N - 4$	3250
$N - 5$	3200
$N - 6$	3000

Using the Least Squares Method, determine the estimated sales revenue for year $N + 1$.

Exercise 4: Continuous Least Squares approximation

Calculate the polynomial $L_1(x)$ of degree less than or equal to 1, which approximates the function $f(x) = e^x$ in the sense of continuous least squares over $[0, 1]$.